Leaderboards Are Not All You Need: A Causal Check-Up for Driving Policies

Anonymous Submission
Paper ID —

Abstract—Choosing which policy to deploy usually starts from its rank on a public leaderboard, narrowed where possible to the scenario type the deployment site cares about — here, scenes where a pedestrian appears. We ask whether that rank is enough to make the choice, and find that it is not: a policy's rank on nuScenes' open-loop metrics or on NAVSIM's leaderboard score, computed on near-pedestrian scenes, predicts its rank at a new deployment site only as well as a coin flip ($\rho = -0.54$–$+0.49$). The reason is that a benchmark score says what a driving policy's trajectories achieved on logged scenes; it does not say whether a vulnerable road user in the corridor had anything to do with that result, and that is why the rank does not travel. We propose a check-up: a short list of counterfactual exams that read a policy's behaviour around pedestrians from a few hundred logged frames, without a simulator, a fleet test, or retraining. Each exam edits real frames—removing the pedestrian from the ego corridor, or re-rendering the frame at night—pairs the edit with a control, and has an independent detector verify it. The central reading is not how far the plan moves but how far it moves away: we decompose the response into its magnitude, the policy's own jitter under an irrelevant edit, and the separation it actually buys from the pedestrian's logged future. On 246 near-pedestrian scenes of a second corpus (NAVSIM, Singapore) across six end-to-end policies, a near-zero response is a reliable certificate that the pedestrian did not enter the decision (it predicts an unchanged outcome in 93–99% of scenes), while a large response is not evidence of avoidance: of the cells in which the plan moved, 43 increased the clearance and 42 decreased it, and only 2.4% of all cells meet all three conditions for genuine avoidance. Moving from left-hand to right-hand drive, what transfers is not how strongly a policy reacts but how steadily it drives: the specificity exam predicts the right-hand-drive rate of sub-1.5 s time-to-proximity at $\rho = -0.94$, beating both the other exams and the leaderboard score computed on that side's own scenes. We fit how each verdict's uncertainty shrinks with sample size, so that a check-up can be priced before it is run: the lighting verdict is settled in 14 frames and exposure in 51, while hazard sensitivity needs hundreds. A pre-registered Boston-to-Singapore test confirms the operating range: orderings and verdicts carry over, point values do not.

I. Introduction

End-to-end driving policies are compared by benchmark scores, and a deployment decision is, implicitly, a bet that the rank those scores produce will still hold once the policy is driving at the new site, not just on someone else's roads. Taking that bet blindly is a bad idea: on scenes where a pedestrian is near the ego corridor, a policy's rank on nuScenes' open-loop metrics or on NAVSIM's leaderboard score predicts its rank at the deployment site only about as well as a coin flip ($\rho = -0.54$–$+0.49$; Section V). The reason is not noise but mechanism: a score is an outcome measure, and an outcome can be right for the wrong reason. A policy that never approaches a pedestrian because it plans short trajectories scores as well as one that sees the pedestrian and yields, and the two behave very differently the first time the scene changes. Telling them apart means asking why the score came out as it did, usually from little data: a closed-loop simulator for the target domain, or a fleet test, is exactly what the decision is meant to save.

We propose a check-up: a short list of exams, each with a control and a known normal range, whose results are read as a diagnosis rather than a ranking (Table I). Every exam is a counterfactual edit of real frames. We remove the pedestrian from the ego corridor, or re-render the frame at night, and read the change in the planned trajectory; an independent detector verifies every edit. The central exam is a hazard-sensitivity curve, measuring how much farther the whole planned trajectory stays from the pedestrian's logged future because the policy can see the pedestrian, against a model-independent hazard level. A specificity term applies a minimal-change principle: when no reaction is needed, the plan should not move.

a) Displacement is not avoidance: What an edit yields most directly—how far the plan moves when the pedestrian is removed—is a poor safety signal on its own, and treating it as one inverts the ranking. We decompose it into the magnitude of the response, the displacement the same policy produces under an edit unrelated to the pedestrian (its own jitter floor), and the separation the response actually buys. On 246 NAVSIM Singapore near-pedestrian scenes, the pedestrian outweighs the irrelevant edit in only 4–23% of scenes; among the cells where the plan did move, 43 moved away from the pedestrian and 42 moved towards it. The asymmetry that survives is one-directional and useful: a near-zero response certifies that the pedestrian did not enter the decision (93–99% of scenes), while a large response certifies nothing.

b) What the six policies do: Applied to six end-to-end policies, the check-up separates quantities a benchmark score merges (Table II). Their plans cover 0.50–2.19 times the distance the human drove. Seeing the pedestrian improves the plan by 0.01–0.07 on a $[-1, 1]$ scale, and in five of six policies the improvement does not grow with the hazard. Collisions with the pedestrian's logged future therefore follow how far a policy plans to drive, not what it perceives. A night rendering moves every policy more than the pedestrian does, and it does so along the interference direction $\hat{v}_I$, which opposes the faithfulness direction $\hat{v}_F$ governing the pedestrian response (Section VIII). We look inside the policies to explain this, although we do not attempt a repair: the pedestrian leaves a trace in the features but is not decoded by the policies' own semantic heads, the planner reads a grid too coarse to locate it, and its plan ends before the pedestrian.

c) What a small check-up can support: A check-up is only useful if it is cheap and if its operating range is known. We fit how each estimate's spread shrinks with the number of frames: the lighting verdict is settled within 14 frames and exposure within 51, while hazard sensitivity and the direction of the lighting effect need hundreds. We then pre-registered seven predictions from a diagnosis made on 88 Boston frames and tested them after the switch to right-hand drive. The ordering predictions transfer ($\rho = 0.90$, against a pre-registered bar of 0.6); the stronger, point-value version of the same claim falls short (Table VI). Rankings and verdicts, not point values, are the unit that survives the domain gap, and for choosing among candidate policies that is enough, so we do not pursue calibration further. Re-measuring the exams on the new side, the one that predicts behaviour there is not the exam measuring how strongly a policy reacts but the one measuring how steadily it drives ($\rho = -0.94$ for specificity); it beats the leaderboard score computed on that side's own full scene set (Section V). The bet named above is therefore answerable: not by trusting a leaderboard rank, but by running a check-up whose predictive exams are known in advance.

d) Contributions:

1) A small-sample check-up for driving policies, built on verified, pixel-level counterfactual edits of real frames rather than the promptable text prompts language-model interpretability usually relies on — to our knowledge among the first uses of concept-vector-style methods on a multi-modal, largely non-promptable input. Its central result is methodological: raw response magnitude is a poor safety signal, and decomposing it into magnitude, jitter and separation turns a two-sided quantity into a one-sided certificate. We use it to diagnose six policies, with an explanation from inside the networks.
2) Evidence, in a new domain, for the linear representation hypothesis: the faithfulness direction $\hat{v}_F$ (response to the pedestrian) and the interference direction $\hat{v}_I$ (response to a change with no driving-relevant information) are geometrically opposed in aggregate (mean cosine -0.54 across four policies), and this opposition is mirrored behaviourally, not just in the representation. The relationship is a property of the pooled data, not a stable per-policy trait (Section VIII), and we report it as such.
3) An account of what small data can and cannot establish: sample-size laws for every exam and a pre-registered left- to right-hand-drive transfer test. The exam that transfers predicts behaviour on the new side better than that side's own leaderboard score does.

TABLE I
The check-up. Each exam, the question it asks, the counterfactual that answers it, and the normal range of a driver that yields.

Exam	Question	Measurement	Normal
Exposure	How far ahead the plan commits	planned / logged distance, 2.5 s	≈1
Hazard sensitivity HS	Does it react to danger	safety gained by seeing the pedestrian, when needed	rises to 1 with hazard
Scaling	Does caution scale with danger	rank corr. of HS with hazard	> 0
Specificity SP	Does it over-react	plan change when no reaction is needed	≈1
Lighting CFR, align	Does it react to noise	night vs. removal, same frame	CFR $\gg 1$, align ≈ 0

II. Related Work

a) Driving policies and their benchmarks: We examine BEV planners of the TransFuser lineage [1] (LTF, DiffusionDrive [2], DiffusionDriveV2 [3]) and vision–language–action models (SimLingo [4], AutoVLA [5], Alpamayo-1.5 [6], [7]). NAVSIM [8], [9] scores their plans by multiplying hard gates with a weighted average of soft terms. Open-loop scores are known to reward proxies [10]. We show that the six-policy spread is carried by a single gate.

b) Behavioural testing: Checklists of targeted behavioural tests [11] and corruption benchmarks [12]–[14] probe models beyond aggregate accuracy. A metric drop, however, does not say whether a policy became more cautious or more reckless. Our exams read signed plan changes against a counterfactual and state how many frames each needs.

c) Causal confusion: Imitation learners can respond to correlates of the expert's action rather than to its causes [15]–[18]. Our exams apply this idea to trained policies: a cause (a pedestrian) and a nuisance (lighting) are intervened on directly.

d) Counterfactual editing and internal representations: Counterfactual explanations edit the scene [19], [20]; we verify every edit with a detector. Steering directions [21] can show that a network encodes what it does not act on [22]; we use them to explain a diagnosis, not as the diagnosis.

III. The Check-Up

A. Counterfactual units

A unit is a logged frame together with an input ego speed. We use the logged speed for every frame, and for frames from approach sequences also counterfactual speeds of 2–8 m/s. For each unit we query the policy three times. The query uses its full input (all cameras, history frames, LiDAR, ego status), edited in one of three ways: unchanged (P), with the target pedestrian removed (Q), or re-rendered at night (N). All policies are read on their common horizon of 2.5 s. Plans and the pedestrian's logged future positions are interpolated to 0.1 s.

a) Removal: The target is located by projecting its annotated 3D box into the camera rather than by taking the dataset's 2D boxes, whose association to the tracked object is not reliable for the short-range, partially occluded pedestrians these scenes contain. It is segmented with SAM [23] (box and torso prompts, clipped to the box, dilated by 9 px) and the mask is filled with LaMa [24]; for the LiDAR policy the points inside the box are deleted as well. The VLA policies read several cameras and we edit only the front one, so we retain only frames in which the target appears in no other view.

b) Night: A closed-form photometric transform (gamma 2.2, gain 0.45, saturation 0.45, blue tint, sensor noise $\sigma = 7$) is applied to every image the policy consumes. The noise seed is fixed per frame.

c) Validity: Every edit is checked by a detector that no policy shares (Faster R-CNN [25], COCO, score ≥ 0.5) on the 248 front-only frames. It recovers the target in 100% of the original images, in 8.5% after removal and in 89.9% after the night rendering (IoU ≥ 0.5). The removal therefore suppresses the pedestrian for an observer outside the policy, while the night rendering leaves it visible and so changes the lighting rather than the scene. Two controls bound the residual artefacts: pasting inpainting patches on empty road puts the artefact at 5–17% of the removal effect at pedestrian scale, and removing off-road pedestrians, which should not matter causally, provides a same-type control.

B. Readouts: the whole trajectory against the pedestrian

Let $c_t(X) = \|X_t - F_t\| - 1.0 - 0.4$ be the clearance at time t between plan X and the pedestrian's logged position F_t, net of a 1.0 m ego half-width and a 0.4 m pedestrian radius. We read four properties of a plan:

- its separation $S(X) = \mathrm{mean}_t \min(c_t, 10\,\mathrm{m})$, i.e., how far the whole trajectory stays from the pedestrian;
- contact $A(X) = \mathbb{1}[\min_t c_t > 0]$;
- safety distance $C(X) = \mathrm{clip}(\min_t c_t / 1\,\mathrm{m}, 0, 1)$;
- time to proximity $T(X)$, the first time $c_t < 1\,\mathrm{m}$, divided by 2.5 s (1 if it never happens).

Every constant above is a physical length or the horizon, not a tuned parameter: the subtracted 1.0 m is the half-width of the nuScenes ego box (4.08×1.85 m), rounded up from 0.93 m, and 0.4 m is the disc a standing adult occupies; the 1 m in C and T is the outer edge of personal space; 2.5 s is the horizon all six policies share; and the 10 m cap keeps scenes where the pedestrian is far away from dominating the mean in S.

These four follow existing conventions rather than new ones. Contact and minimum clearance are the collision and margin measures of the surrogate-safety literature, and time to proximity is an encroachment-time variant of the same family [26], [27]; reading distance to a person over the whole trajectory, rather than the ego's own speed, is how navigation near people is evaluated in robotics [28], and the 1 m threshold in C and T is the outer edge of personal space [29]. Speed changes are deliberately not a readout: a plan is judged by where it goes relative to the pedestrian, not by how it decelerates, since the same deceleration is safe or unsafe depending on where the person is.

C. Exams

a) Exposure: The ratio of planned to logged distance over 2.5 s, on frames where the human moved at least 0.5 m.

b) Hazard sensitivity: A unit needs a reaction if the pedestrian-blind plan comes within 1 m of the pedestrian, $C(Q) < 1$. On such units,

$$\mathrm{HS} = \tfrac{1}{4}\left[\Delta A + \Delta C + \Delta T + \tanh(\Delta S / 1\,\mathrm{m})\right] \in [-1, 1], \quad (1)$$

with $\Delta = (\cdot)(P) - (\cdot)(Q)$: the safety that seeing the pedestrian buys. The composite is ours, but neither of its two ingredients is new. Its four terms are the standard surrogate safety measures—contact, minimum clearance, time to proximity and separation [26], [27]—and averaging normalised sub-scores into one index is how the benchmark we compare against builds EPDMS [8], so the two remain commensurable. What the check-up adds is the paired query: each sub-score is read as a difference between the same policy with and without the pedestrian, which is the ablation form used to attribute behaviour to an input in interpretability work [21], [30]. Each term is bounded in $[0, 1]$, so their mean is bounded in $[-1, 1]$; tanh squashes the one unbounded term at a 1 m scale. We plot HS against the model-independent hazard level $a_{\mathrm{req}} = v^2 / 2\max(d - 2, 0.5)$, the deceleration needed to stop before a pedestrian at distance d—the deceleration-rate-to-avoid-the-crash of the surrogate-safety literature [26]—and alternatively against the time-based $\mathrm{TTC}_0 = (d - 2)/v$ [27]. Units with input speed below 1 m/s are excluded, because a stationary ego cannot move toward or away from anyone.

c) Scaling and specificity: Scaling is the rank correlation of HS with a_{req} across needed units. Specificity applies the minimal-change principle to the units that need no reaction: $\mathrm{SP} = \exp(-\bar{D}(P, Q)/0.5\,\mathrm{m})$, where $\bar{D}$ is the mean displacement between the two plans.

d) Lighting: For the same frames, $\mathrm{CFR} = \mathbb{E}\|P - Q\| / \mathbb{E}\|P - N\|$ compares the plan change caused by removing the pedestrian with that caused by the night rendering. Alignment is the median cosine between $N - P$ and $Q - P$. A pure policy has $\mathrm{CFR} \gg 1$ and alignment near zero. Positive alignment means the night pushes the plan the way removing the pedestrian does. In Section VIII we read

$Q - P$ and $N - P$ as two representation-space directions, $\hat{v}_F$ (the faithfulness direction, the pedestrian response) and $\hat{v}_I$ (the interference direction, the response to a change with no driving-relevant information); alignment is their cosine.

D. Policies and frames

We examine LTF and DiffusionDrive (front camera and ego status), DiffusionDriveV2 (camera and LiDAR), SimLingo (InternVL2-1B), AutoVLA (Qwen2.5-VL-3B; three cameras $\times$ four frames) and Alpamayo-1.5 (four cameras; a diffusion head over the VLM cache). All use released weights. The frames are nuScenes [31] keyframes. They cover corridor pedestrians (lateral clearance to the ego corridor ≤ 1 m) at 5–12 m, their off-road counterparts, and 30 approach sequences at 5.8–14.6 m with logged pedestrian futures. The near-pedestrian units come from 236 front-only frames, 88 in Boston (left-hand drive) and 148 in Singapore (right-hand drive). Alpamayo-1.5 covers 170 of the 236, since it needs 1.6 s of history and 6.4 s of future within the scene. Section VII repeats the same two interventions on a second, right-hand-drive corpus, the NAVSIM test split [8], where 254 Singapore scenes place a pedestrian near the ego corridor and 246 of these also carry a complete logged pedestrian future. It is used only for the transfer and decomposition results; Table II is measured on the nuScenes frames alone. We fix DiffusionDrive's diffusion seed so that paired queries share their noise.

IV. Diagnosis of Six Policies

A. The report

Table II and Fig. 1c give the six profiles.

a) Exposure: The six policies do not even agree on how far to plan, and this alone will turn out to explain more of the collision picture below than perception does: exposure spans a factor of four, from DiffusionDrive at half the human's distance to SimLingo at more than twice it, with AutoVLA and Alpamayo-1.5 near the human's own.

b) Hazard sensitivity: No policy manages the pedestrian risk. On units that need a reaction, seeing the pedestrian buys 0.01–0.07 of safety on a $[-1, 1]$ scale. The benefit does not grow with the hazard: its rank correlation with a_{req} is negative for five policies, down to -0.51 for Alpamayo-1.5. DiffusionDriveV2 is the only exception ($+0.29$), and its benefit stays below 0.15. The time-based hazard axis gives the same picture (Fig. 2b). The benefit is also unreliable in direction: counting only cases where seeing the pedestrian changes the outcome, it turns a collision into a clean pass about as often as it turns a pass into a collision, for every policy (DiffusionDrive 9 vs. 5, SimLingo 18 vs. 30, the rest at most 5 either way; all McNemar $p \geq 0.11$) — what reads as a safety margin in Table II is, case by case, close to a coin flip.

c) Specificity: A policy that moves when nothing requires it is not merely cautious, it is unpredictable, and unpredictability is what turns out to travel across a domain shift (Section VI). SimLingo's plan moves most when nothing requires it (SP = 0.42); the other five stay steadier, at 0.67–0.79.

d) Collisions: Collisions with the pedestrian's logged future follow how far a policy plans to drive, not what it perceives. At the logged speed, the visible and removed rates differ by at most 0.4 points for every policy, and at 8 m/s by at most 3.8. SimLingo, which plans the longest trajectories, collides most. The low-exposure policies almost never reach the pedestrian. DiffusionDrive is safe because it stays short, AutoVLA and Alpamayo-1.5 cover a human-like distance without yielding, and SimLingo covers the most ground with the least stable plan.

B. Looking inside: why the risk goes unmanaged

The check-up only diagnoses; the following explains the diagnosis for the three BEV planners (Table III, Fig. 2).

a) The pedestrian leaves a trace but is not decoded: The planners' own BEV semantic heads do not localise the pedestrian. On 520 NAVSIM near-pedestrian scenes, the pedestrian channel is no brighter at the pedestrian than at a mirrored empty location (Table III, top). It is 20–55 times weaker than the vehicle channel and never wins the argmax inside a pedestrian's box. The pedestrian does leave a trace in the features, however. The direction between SimLingo's representations with and without the pedestrian is steerable: it shortens the plan by 2.9 m at the largest dose, while random directions lengthen it. Location-specific information also survives in the 1 m side branch, as shown next.

b) The planner cannot locate it: The trajectory head reads an 8×8 grid at 8 m per cell. In 73% of 94 pairs of corridor and off-road pedestrians, the two fall in the same or an adjacent cell. A corridor-minus-off-road direction injected there does not beat random directions ($p = 0.27$). At the 1 m side branch it does (52-fold, $p = 0.007$), but with $\sim 1/460$ of the effect on the trajectory; in LTF that branch does not reach the trajectory at all.

c) The plan does not reach the pedestrian: Plans end 5–9 m ahead, while the median corridor pedestrian stands 20 m away; only 5–11% of pedestrians are within reach. A distance sweep over 850 nuScenes scenes (974 pedestrian records) confirms that the limit is geometric rather than perceptual: once the pedestrian is within 12 m, LTF begins to respond, and at 8–12 m its plan changes by 0.23 m ($p = 5 \times 10^{-7}$), which separates corridor from off-road pedestrians after mask area is regressed out (AUC 0.65; 0.67 on an independent draw). Raising the input speed removes that response again, and collisions then rise identically with and without the pedestrian in view (Fig. 2a). DiffusionDrive never responds at any distance.

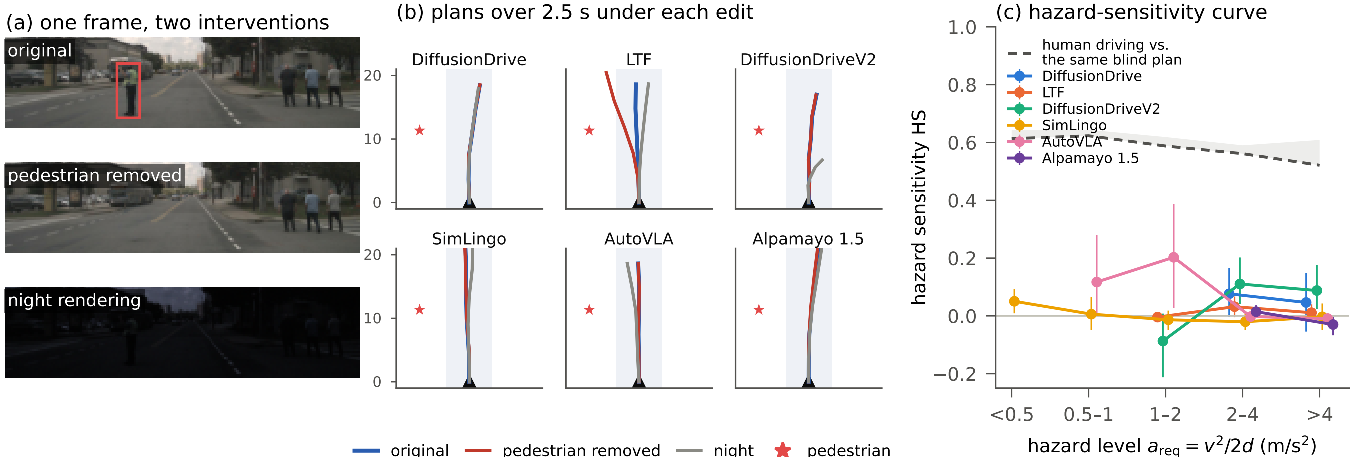

Fig. 1. One frame, two edits, six policies, and the resulting curve. (a) A worker in a high-visibility vest stands at the edge of the ego corridor 9.6 m ahead; the ego vehicle moves at 7.6 m/s. The detector finds the worker in the night image but not in the removed one. (b) Plans over 2.5 s under each input: original (blue), pedestrian removed (red), night (grey); the star marks the worker. (c) Hazard-sensitivity curves over all near-pedestrian units that need a reaction (mean, 95% CI, bins with ≥ 8 units; points are dodged horizontally within each bin so the intervals do not overlap, and a policy appears only where it has enough units, which is why the curves start at different hazard levels). HS is a dimensionless index in $[-1, 1]$. The dashed line scores the trajectory the human actually drove against the same pedestrian-blind plan, on the same four readouts; the shaded strip extends it to a plan that keeps full distance. It is a gap, not a human HS: the logged driver has no ablated counterpart, so no subtrahend of its own. Real driving is 0.5–0.6 safer than the blind plans these policies produce, and seeing the pedestrian recovers almost none of that gap.

TABLE II

The check-up report of six policies (pooled near-pedestrian units, both driving sides; 95% bootstrap intervals over frames). Exposure: planned / logged distance over 2.5 s (1 = human). HS: safety gained by seeing the pedestrian when it matters, on $[-1, 1]$ (a yielding driver approaches 1). Scaling: rank correlation of HS with hazard level a_{req} (does it respond more when risk is higher). SP: how little the plan moves when no reaction is needed (over-reaction; 1 = unmoved). Collision: contact with the pedestrian's logged future, pedestrian visible / removed, at the logged speed and at 8 m/s. Right: the NAVSIM benchmark on 783 Singapore scenes.

	exposure	HS	scaling	SP	collision (%), visible / removed		NAVSIM	
Policy	(distance)	(yielding)	(scaling)	(over-reaction)	logged v	8 m/s	EPDMS	DAC
DiffusionDrive	0.50 [0.46,0.55]	+0.07 [-0.01,0.14]	−0.25	0.78 [0.75,0.81]	0.0 / 0.0	7.6 / 9.2	0.535	0.673
LTF	0.79 [0.75,0.84]	+0.02 [-0.01,0.04]	−0.03	0.79 [0.76,0.81]	0.4 / 0.4	9.9 / 10.7	0.509	0.630
DiffusionDriveV2	0.88 [0.77,0.95]	+0.06 [0.01,0.11]	+0.29	0.67 [0.63,0.71]	0.4 / 0.0	3.1 / 6.1	0.652	0.820
SimLingo	2.19 [1.98,2.65]	+0.01 [-0.02,0.03]	−0.19	0.42 [0.36,0.48]	17.4 / 17.4	40.5 / 36.6	0.359	0.452
AutoVLA	1.01 [0.97,1.11]	+0.02 [-0.00,0.05]	−0.18	0.76 [0.72,0.79]	1.7 / 1.3	32.1 / 29.8	0.683	0.814
Alpamayo 1.5	1.13 [1.06,1.21]	+0.01 [-0.01,0.02]	−0.23	0.84 [0.81,0.87]	0.0 / 0.0	11.2 / 11.2	0.663	0.789

C. What the benchmark sees

On 783 NAVSIM scenes scored identically, 95% of the between-policy variance of EPDMS comes from the two map-compliance gates (drivable area 76%, driving direction 19%); the no-collision gate contributes 1% (Table II). The benchmark ranks these policies by map compliance, and none of the exams above is visible in it.

V. Against the Standard Evaluations

How does the check-up relate to the evaluations these policies are usually judged by? We ran both on the same six policies (Table IV).

a) nuScenes open-loop metrics: On our 236 near-pedestrian frames, removing the pedestrian changes the L2 error by at most 0.07 m, frontal collisions by at most 1.7 points and pedestrian collisions by at most 3.0. The metric cannot tell whether a policy saw the pedestrian. Its ordering follows how far a policy's planned distance departs from the human's (rank correlation +0.66 with $|\text{exposure} - 1|$): L2 rewards driving at the human's speed. We exclude rear-end contacts from the collision count. The logged traffic does not react, so a policy that plans to drive slower than the human is scored as being hit from behind.

b) NAVSIM, sliced to near pedestrians: On all 783 Singapore scenes, AutoVLA, Alpamayo-1.5 and DiffusionDriveV2 score highest. On the 260 scenes with a pedestrian near the ego corridor, the ordering is unrelated to that one ($\rho = -0.09$). The two rank columns of Table IV can be read side by side: the leaderboard's first (AutoVLA) falls to fifth and its third (DiffusionDriveV2) to last, while its fourth and fifth (DiffusionDrive, LTF) rise to second

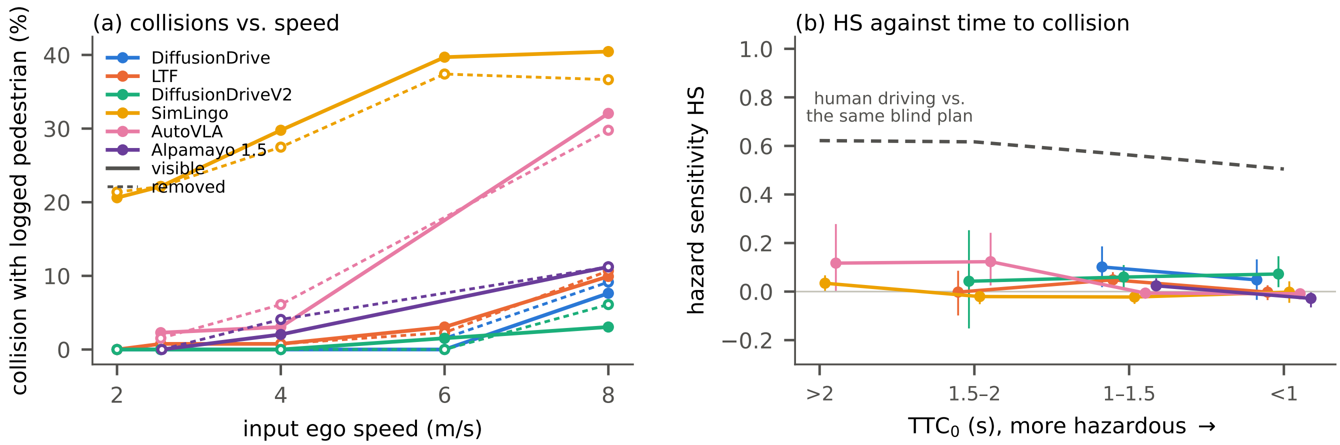

Fig. 2. What the unmanaged risk costs. (a) Collision with the pedestrian's logged future along the plan, at the logged speed and at counterfactual input speeds; solid: pedestrian visible, dashed: removed. Each pair of curves nearly coincides, and every curve rises with speed. (b) Hazard sensitivity against the time-based hazard axis TTC_0 (bins with ≥ 8 units); colours as in (a), dashed: the same human reference as Fig. 1c.

TABLE III
Not decoded, not transmitted, and out of reach (BEV planners). Top: the policy's own BEV semantic head on 520 NAVSIM near-pedestrian scenes (in parentheses: the same readout at a mirrored empty location). Middle: difference-of-differences injection (corridor minus off-road pedestrian direction), reported as effect over the median of random directions, with permutation p. Bottom: planning reach.

	DD	LTF	DDv2
Ped. peak at pedestrian / global	0.07 (0.11)	0.07 (0.07)	0.09 (0.15)
Ped.-box px won by ped.	0%	0%	0%
Pedestrian / vehicle peak	0.018	0.023	0.052
Injection, 8 m grid (planner)	2.9× (p=.27)	p=1.0	–
Injection, 1 m grid (branch)	52× (p=.007)	≡ 0 (14/14)	–
effect vs. 8 m grid	∼1/460	–	–
Median plan length (m)	5.3	7.1	8.9
Median pedestrian distance (m)	20.8	20.8	20.6
Pedestrian within plan reach	5%	11%	10%

and third. Whichever policy a leaderboard reader would have picked, the near-pedestrian scenes would have picked another. In these scenes the no-collision gate carries 100% of the spread, against 1% over all scenes. The cause is a trait the check-up had not examined: 63% of these scenes show a human driver waiting. DiffusionDriveV2 and AutoVLA plan to move off and run into the stopped traffic. On our own frames where the human waits, DiffusionDriveV2 plans to move off in 81% of cases, against 3% for DiffusionDrive and 0% for Alpamayo-1.5. This agrees with NAVSIM for the extremes but not for AutoVLA. Of our exams, specificity predicts the near-pedestrian ordering ($\rho = +0.66$), while hazard sensitivity does not (+0.14), because no policy has any.

c) Pedestrian TTC across driving sides: Following the plan against the pedestrian's logged future, we count the moving-ego scenes in which the minimum time to collision falls below 1.5 s. Near-pedestrian scenes from the left-hand-drive cities (nuScenes Boston and NAVSIM Las Vegas, Boston, Pittsburgh) and from Singapore give orderings that agree at $\rho = +0.77$. Both orderings coincide with planned distance ($\rho = +1.00$ and +0.77 with exposure). They agree only partly with hazard sensitivity (+0.83, +0.49), which is near zero for every policy, and not with lighting invariance (+0.14, −0.43). NAVSIM's own TTC term, which counts all objects over 4 s, agrees only weakly (0.31–0.54).

d) Which evaluation predicts behaviour in another domain?: This is the question a deployment decision has to answer, so we pose it directly in Table V. The target is the ordering of the six policies on the Singapore near-pedestrian scenes (original images), by TTC violations, clearance to the pedestrian, pedestrian collisions and EPDMS. Each predictor is an ordering obtained without that data. The standard evaluations do not predict it: nuScenes L2 and collision rates, and the NAVSIM leaderboard score, correlate with it at −0.54 to +0.49. The same pedestrian-centred readouts taken on Boston or on the US cities do predict it (0.83–1.00), and so does the check-up's exposure exam (0.94). Near-pedestrian EPDMS is predicted best by near-pedestrian EPDMS elsewhere (0.89) and by specificity (0.77). Hazard sensitivity and lighting invariance measure something else, and they do not predict these orderings.

e) Reading: Where a standard metric measures what an exam measures, they agree (pedestrian collisions: $\rho = +0.94$; TTC and exposure). None shows whether the pedestrian changed the plan, and the full-set score hides

TABLE IV
The same six policies under the standard evaluations. Left: nuScenes open-loop metrics on the same 236 near-pedestrian frames, original / pedestrian removed (L2 to the human trajectory averaged over 0.5–2.5 s; collisions with objects ahead of the ego, rear-end contacts by non-reactive logged agents excluded). Middle: NAVSIM EPDMS on all 783 Singapore scenes and on 260 Singapore scenes with a pedestrian within 1 m of the ego corridor and 20 m ahead; the leaderboard ordering does not survive the slice. Right: pedestrian TTC along the plan (share of moving-ego scenes with minimum TTC < 1.5 s, lower is better) on left-hand-drive near-pedestrian scenes (nuScenes Boston + NAVSIM Las Vegas/Boston/Pittsburgh) and right-hand-drive ones (NAVSIM Singapore), with the resulting rank.

	nuScenes open-loop (orig. / removed)			NAVSIM EPDMS (rank)		pedestrian TTC < 1.5 s (rank)	
Policy	L2 (m)	collision (%)	ped. coll. (%)	all	near ped.	left-hand	right-hand
DiffusionDrive	0.98 / 0.99	14.0 / 14.8	1.3 / 1.7	0.535 (4)	0.744 (2)	3.2 (1)	14.1 (1)
LTF	0.88 / 0.91	4.2 / 3.8	2.5 / 1.7	0.509 (5)	0.735 (3)	5.4 (2)	16.5 (2)
DiffusionDriveV2	1.35 / 1.31	6.4 / 7.2	3.8 / 3.8	0.652 (3)	0.367 (6)	18.3 (3)	22.4 (5)
SimLingo	3.28 / 3.35	37.3 / 35.6	22.5 / 19.5	0.359 (6)	0.583 (4)	61.3 (6)	56.5 (6)
AutoVLA	0.68 / 0.66	6.4 / 5.9	3.8 / 3.4	0.683 (1)	0.413 (5)	19.4 (4)	21.2 (4)
Alpamayo 1.5	0.77 / 0.72	2.4 / 2.4	0.6 / 0.6	0.663 (2)	0.760 (1)	25.4 (5)	20.0 (3)

TABLE V
Which evaluation predicts behaviour near pedestrians in another domain? Rank correlation between the six-policy ordering given by each evaluation (none uses the target data) and the ordering observed on 260 NAVSIM Singapore scenes with a pedestrian near the ego corridor (original images, moving ego). Bold: $|\rho| \geq 0.89$ (two-sided 5% for n=6).

	Singapore near-pedestrian ordering by			
Ordering from	TTC	clearance	collisions	EPDMS
Same readouts, nuScenes Boston				
TTC violations	+0.94	+0.83	+0.94	+0.14
clearance	+0.83	+0.94	+0.83	+0.60
collisions	+0.83	+0.94	+1.00	+0.09
Same readouts, NAVSIM US cities				
TTC violations	+0.83	+0.94	+1.00	+0.09
clearance	+0.83	+0.71	+0.89	−0.03
near-ped. EPDMS	+0.77	+0.54	+0.43	+0.89
nuScenes open-loop (Boston)				
L2	+0.43	+0.09	+0.03	+0.49
collision	+0.26	−0.09	−0.03	+0.03
ped. collision	+0.60	+0.37	+0.26	+0.71
NAVSIM leaderboard				
EPDMS, 783 scenes	+0.14	−0.09	−0.03	−0.09
Check-up (Boston)				
exposure	+0.83	+0.94	+1.00	+0.09
hazard sensitivity	−0.66	−0.89	−0.94	+0.09
specificity	+0.83	+0.71	+0.60	+0.77
lighting CFR	−0.20	−0.09	+0.14	−0.94
patience	+0.31	−0.03	−0.14	+0.77

the near-pedestrian scenes.

VI. How Much Data Does a Check-Up Need?

A. Fitting

For every exam and policy we subsample n of the 170 frames that all six policies share, 400 times, and fit the spread of the estimate with $\sigma(n) = a\sqrt{1/n - 1/N}$ (Fig. 3).

a) Well-fitting exams: Exposure, the lighting ratio and specificity obey this law ($R^2 = 0.97$–1.00). The fitted a converts directly into the number of frames a verdict needs.

- Lighting outweighs the pedestrian (CFR $<$ 1) is settled for every policy within 14 frames.
- Exposure differs from the human's is settled within 51 frames for the four policies whose exposure is far from one. For AutoVLA and Alpamayo-1.5 it would take hundreds to thousands of frames; this is itself a finding, since they plan a human-like distance.
- Orderings of the six policies stabilise quickly: exposure with 10 frames (96% of subsamples reach $\rho \geq 0.6$ with the full ordering) and specificity with 40 (99%).

b) Poorly fitting exams: Hazard sensitivity and the direction of the lighting effect behave differently. Their fits are poor (R^2 as low as 0.26 and -0.09) wherever few units need a reaction. Deciding that HS $\neq 0$ needs 57–1,072 frames, and the ordering reaches 84% stability only at 90 frames. The ordering by alignment is unstable even at 130 (72%).

The division is therefore clear. A few dozen frames diagnose planned distance, over-reaction and sensitivity to lighting. Whether a policy manages hazard, and in which direction lighting pushes it, needs hundreds of frames that contain a genuine need to react.

B. A pre-registered transfer test

Does a diagnosis made on one driving side describe the same policy on the other? Before computing any right-hand-drive statistic, we fixed seven predictions and their decision rules. They were derived from the profile of the 88 Boston frames, and we tested them on the near-pedestrian units of the 148 Singapore frames (Table VI). The registration covered the six policies public at that time. NVIDIA subsequently released Alpamayo 1.5, which accepts navigation instructions, and we use it in place of Alpamayo-R1 throughout this paper; because both driving sides of Alpamayo 1.5 were measured together, its numbers do not satisfy the registration discipline, so the table is evaluated on the five policies that do. Retaining the replaced policy changes none of the seven verdicts.

a) What transferred: 4 of the seven predictions hold, and they are the orderings and verdicts. The night rendering outweighs the pedestrian in every policy. Exposure keeps its order ($\rho = 0.90$), and so does over-reaction, with

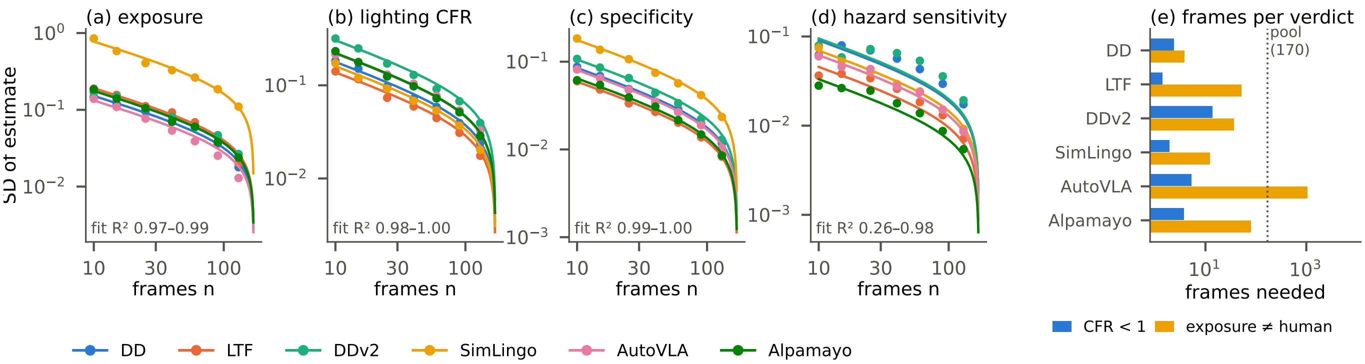

Fig. 3. Fitting estimator spread against sample size. (a–d) Points: standard deviation of each exam's estimate over 400 subsamples of n frames, drawn without replacement from the 170 frames all six policies share. Lines: the fitted finite-population law $a\sqrt{1/n - 1/N}$. Exposure, lighting and specificity follow it closely. Hazard sensitivity fits poorly for policies with few units that need a reaction. (e) Frames needed to place the estimate two standard deviations from the verdict threshold, extrapolated from the fit. For AutoVLA and Alpamayo-1.5, exposure is indistinguishable from the human's.

TABLE VI

Pre-registered transfer test. Diagnoses made on 88 left-hand-drive frames (Boston) predict the near-pedestrian units of 148 right-hand-drive frames (Singapore). Predictions and decision rules were fixed before any right-hand statistic was computed. Evaluated on the five policies whose registration the Alpamayo-1.5 replacement leaves intact (Section VI); retaining the replaced policy changes no verdict.

	Prediction	Holds	Evidence
P1	Lighting outweighs the pedestrian (CFR < 1) for every policy	yes	max upper CI 0.70
P3	Exposure ordering transfers ($\rho \geq 0.6$; SimLingo top, DD bottom)	yes	$\rho = 0.90$
P4	Over-reaction ordering transfers ($\rho \geq 0.6$; SimLingo lowest SP)	yes	$\rho = 0.90$
P7	SimLingo collides most; visible vs. removed within 3 pts	yes	gap 2.7 pts
P2	Hazard sensitivity stays below 0.15 (upper CI)	no	fails for DD, DDv2
P5	Point values stay within the left-hand CI ($\geq$70% of 15 cells)	no	60%
P6	Purer policies (lower |align|) shift less	no	$\rho = -0.90$ (opposite)

SimLingo the most unstable. SimLingo collides most, with the visible and removed rates within 3 points of each other.

b) What did not: Three predictions fail.

- Point values drift: only 60% of the model–exam cells stay inside their Boston interval.
- In Singapore, DiffusionDrive and DiffusionDriveV2 show a small but non-zero hazard sensitivity (HS = 0.09 and 0.08).
- The "purity" hypothesis is reversed. We had predicted that policies whose lighting effect is more orthogonal to their pedestrian response would shift less between sides; the observed rank correlation is -0.90. Alignment is itself unstable between the two sides (Section VIII).

TABLE VII

Re-measuring the two axes in the new domain. CFR measured on Boston frames, on Singapore frames, and per scene on the ten Singapore cases (median). The verdict CFR < 1 holds on both sides for every policy, while the value drifts by 0.6–1.7×; the across-policy ordering of CFR agrees between sides only at $\rho = +0.37$ (specificity: $\rho = +0.83$). Right: share of near-pedestrian scenes in which each rule holds (P-F: a near-zero F axis implies removing the pedestrian leaves clearance and contact unchanged; P-I: CFR < 1 implies the night rendering moves the outcome at least as much as the removal). Both rules are two-sided: each cell gives the hit rate when the premise holds / when it is reversed (fwd/rev), so every scene receives a prediction. and the share of scenes whose minimum time-to-proximity to the pedestrian's logged future falls below 1.5 s. The large-sample columns use the 246 NAVSIM Singapore scenes all six policies share; the Boston column is measured on the nuScenes frames of Table II.

	CFR			TTC	rule holds %	
Policy	Bos.	Sing.	scene	$<$1.5 s	P-F	P-I
DD	0.17	0.24	0.03	13%	95/74	95/97
LTF	0.34	0.19	0.12	16%	96/100	97/100
DDv2	0.39	0.52	0.13	22%	96/67	96/87
SimLingo	0.32	0.19	0.06	56%	93/73	95/91
AutoVLA	0.41	0.46	0.00	21%	95/60	98/86
Alpamayo 1.5	0.15	0.26	0.13	20%	99/33	94/93

c) Reading: A small check-up can therefore report an ordering and a verdict for use in a new domain, but not a calibrated value.

VII. Re-measuring F and I in the New Domain

Section VI moved a diagnosis from one driving side to the other. The more useful question for deployment is whether the method moves: can a handful of counterfactual cases in the new domain characterise how the policies behave there? We repeat both interventions on the 246 NAVSIM Singapore near-pedestrian scenes and read, per scene, the two behavioural displacements that correspond to $\hat{v}_F$ and $\hat{v}_I$ at the output: F, the plan displacement caused by removing the target pedestrian, and I, that caused by the night rendering.

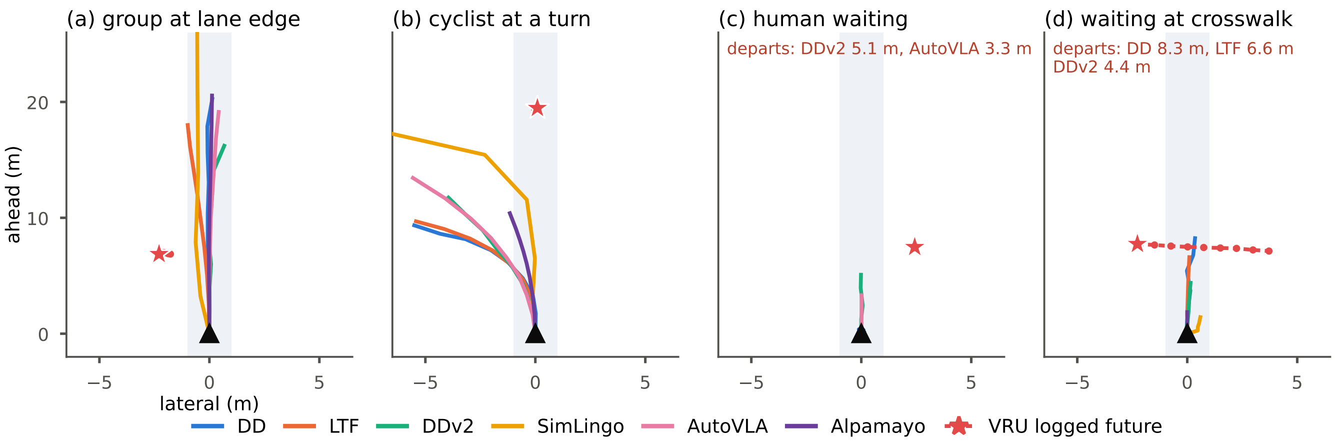

Fig. 4. Four of the ten Singapore cases, against predictions made on Boston frames. Ego at the origin, its corridor shaded, the VRU's logged future dashed from the star. (a) A group at the lane edge: every policy drives past within 0.3–0.9 m and none yields—the exposure and hazard-sensitivity rows of Table II in a single frame. (b) A cyclist at a left turn: SimLingo, the policy the check-up marks as planning the longest distance, is the one that comes closest (3.2 m, 0.7 s to proximity), while the short planners stay 6.7–10 m away. (c) A human waiting behind stopped traffic: DiffusionDriveV2 departs, as its 81% Boston departure rate predicts; AutoVLA departs too, which its 9% rate does not. (d) A human waiting at a crosswalk: DiffusionDrive plans 8.3 m across it, the clearest miss of the patience exam (Boston rate 2.8%). Over the ten cases the scene-level predictions hold in 41 of 60 policy–scene cells: the check-up orders policies well and calls individual scenes only two times in three.

a) What F and I say per scene: Across the ten detailed cases and six policies (60 cells), the median F is 0.05 m and the median I is 0.62 m; the lighting moves the plan more in 85% of cells. Two rules follow from the axes. We state each in both directions, so that every cell receives a prediction:

- P-F: if $F < 0.5$ m, removing the pedestrian should leave the clearance and the contact outcome unchanged; if $F \geq 0.5$ m, it should change them.
- P-I: if CFR < 1, the night rendering should move the outcome at least as much as the removal; if CFR ≥ 1, the removal should move it at least as much as the night.

The two directions do not fare equally. On the detailed cases, P-F holds in 54/55 cells in the forward direction but only 2/5 in the reverse; P-I holds in 51/51 and 8/8. Scaling both interventions to the full pool, which all six policies cover, reproduces the asymmetry exactly (Table VII): P-F holds in 93–99% of scenes forward against 33–100% reversed, while P-I holds in 94–98% and 86–100%. A near-zero F is therefore a reliable certificate that the pedestrian is not being used, but a large F is not a promise of safer behaviour.

b) When the plan does move, which way does it move?: The reverse direction fails for an instructive reason: F measures how far a plan moves, not where it moves to. We therefore decompose it (Fig. 5), reading F against I—the same policy's jitter floor under an unrelated edit—and against ΔS, the change in clearance to the pedestrian's logged future. A response counts as avoidance only if all three hold: the plan moved ($F \geq 0.5$ m), it moved more than the policy's own jitter ($F > I$), and the movement bought distance ($\Delta S \geq 0.5$ m).

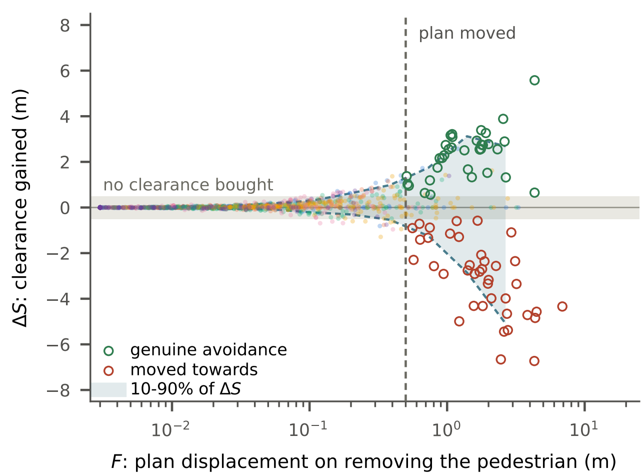

Fig. 5. Displacement is not avoidance. Every policy–scene cell of the 246 Singapore scenes: the plan displacement caused by removing the pedestrian (F, log scale) against the clearance it buys (ΔS). The band is the 10–90% spread of ΔS per F bin. Below the $F = 0.5$ m threshold it is pinched at zero; above it, it opens almost symmetrically—a plan that moves is as likely to move towards the pedestrian as away from it. Circled: cells meeting all three conditions for genuine avoidance (green) and cells whose plan moved towards the pedestrian (red).

Three readings follow. First, the pedestrian outweighs the irrelevant edit in only 4–23% of scenes, so most of what F registers sits inside the policy's own variability—the pinched band of Fig. 5. Second, only 35 of 1476 cells (2.4%)

are genuine avoidance. Third, the displacement that does occur is not aimed: of the 121 cells in which the plan moved, 43 increased the clearance and 42 decreased it. Per policy the split differs sharply: DiffusionDrive reacts in few scenes but in the right direction (median $\Delta S = +2.16\,\mathrm{m}$), while SimLingo moves in more scenes than the other five combined and moves towards the pedestrian more often than away (32 against 13, median -0.59 m); it also has the highest rate of sub-1.5 s time-to-proximity (Table VII). Finally, what a policy yields does not grow with the hazard: the rank correlation between ΔS and the deceleration the scene demands is -0.37–$+0.07$.

An exam that rewarded responsiveness alone would therefore rank the least safe policy first.

c) What a small sample in the new domain buys: We draw k scenes at random from the pool, re-measure F and I on them and compare against the large-sample values, 200 draws per k. Ordering and level behave very differently. The ordering of how dangerously the policies behave—the share of scenes whose minimum time-to-proximity falls below 1.5 s—is recovered almost immediately: $\rho = +0.64$ at $k = 10$ and $\rho = +0.65$ at $k = 20$ ($\rho \geq 0.6$ in 70% of draws), rising to $\rho = +0.76$ at $k = 40$. The level is not: the same $k = 20$ sample misstates the violation rate by 13 points and CFR by 0.116; the rule hit-rates sit in between, at 3.6 points. Importing the Boston CFR values instead misstates the Singapore ones by 0.099, already worse than re-measuring $k = 40$ fresh scenes (0.078), while 80 scenes reach 0.054 and 160 0.027. Values measured elsewhere are worth about forty scenes measured here.

d) Which exam travels: Of the exams in the report, the one that predicts the new domain's behaviour is not the one that measures how much a policy reacts, but the one that measures how steadily it drives. Correlating each Boston exam against the Singapore rate of sub-1.5 s time-to-proximity over the six policies, specificity gives $\rho = -0.94$—the only value significant at the two-sided 5% level for n=6—against +0.77 for exposure and +0.37 for CFR. Specificity asks how far a plan moves when an irrelevant part of the scene is edited; a policy that answers "not at all" drives steadily, and steadiness is what survives the change of side. (Exposure travels too: on the larger original-image set of Section V it reaches 0.94 against the same target. Specificity is the stronger predictor and the only one that clears significance here.) The decomposition above says why: displacement that is not aimed away from the pedestrian is not protection, and a policy producing a lot of it is dangerous on both sides.

e) What transfers and what must be re-measured: The verdict CFR < 1 holds on both sides for every policy, but the value drifts by a factor of 0.6 to 1.7 and its across-policy ordering barely agrees between sides ($\rho = +0.37$, against +0.83 for specificity); per-scene medians are weaker still (Table VII). The instrument transfers while its readings do not.

TABLE VIII
What the night rendering does to the plan. CFR: plan change from removing the pedestrian over plan change from the night rendering, same frames (corridor pedestrians 5–12 m, 95% CI; ideal $\gg 1$). Speed: change of planned mean speed over 2.5 s under the night rendering. ΔS: change of whole-plan separation from the pedestrian (negative = closer). Both on frames where the ego moves ($\geq$1 m/s). Align: direction of the night-induced change relative to the removal-induced one, Boston → Singapore (0 = orthogonal). Bold: $p < 0.01$.

Policy	CFR [95% CI]	speed	ΔS (m)	align
DD	0.30 [0.20,0.43]	**+37%**	**−0.37**	+0.55 → +0.09
LTF	0.36 [0.24,0.49]	**+12%**	**−0.24**	−0.09 → −0.42
DDv2	0.49 [0.33,0.70]	**−8%**	**+0.15**	+0.63 → +0.16
SimLingo	0.31 [0.21,0.44]	**+16%**	**−0.47**	+0.30 → +0.14
AutoVLA	0.38 [0.21,0.57]	0%	0	+0.65 → +0.62
Alpamayo 1.5	0.18 [0.10,0.30]	−2%	+0.03	+0.81 → −0.30

VIII. Lighting: Magnitude, Direction, and the Pedestrian

Lighting should not change what a driver does, so this exam asks how far a policy is moved by an irrelevant change. We analyse it last. It is the most robust result in the check-up and also the one whose direction is least stable.

a) Magnitude: On the same frames, removing a pedestrian from the corridor moves every policy's plan only 0.18–0.49 times as far as the night rendering does (Table VIII). Every 95% interval lies below one. The result holds on detector-verified edits (CFR = 0.30–0.64), where only Alpamayo-1.5's interval (45 frames) reaches above one. Tracing the rendering on a separate pool of 312 lead-vehicle events shows that the effect is carried by sensor noise rather than darkness. For DiffusionDrive, darkening alone raises the planned speed by +0.08 m/s and darkening with noise by +2.05 m/s. The effect is also global: perturbing only the sky, with the corridor untouched, recovers 19% of the effect of deleting the lead vehicle.

b) Direction: faster, and therefore closer: Darkness reduces visibility, so a policy that slows at night is doing something defensible; one that speeds up is not. On frames where the ego moves, three policies plan to go faster under the night rendering (Table VIII): DiffusionDrive by 37%, SimLingo by 16% and LTF by 12%. The gain grows along the horizon; for DiffusionDrive it rises from +0.12 m/s in the first half-second to +1.03 m/s in the last. As a result, their whole plans pass 0.24–0.47 m closer to the pedestrian. The approach is longitudinal. The more the plan's endpoint moves forward, the closer it gets (rank correlation −0.73 to −0.96), whereas the endpoint shifts toward the pedestrian's side in no more than 47% of frames, which is no more often than chance. Contacts barely change, because most pedestrians remain beyond the 2.5 s plan. DiffusionDriveV2, the only policy that also reads LiDAR, is the one that slows at night (−8%) and keeps a larger distance. AutoVLA and Alpamayo-1.5 are unaffected. In representation space the night direction opposes the hazard direction (mean cosine -0.54 in four policies), so the policies encode "it got dark" as they encode "the hazard is gone". We write $\hat{v}_F$ for the faithfulness direction (response to the pedestrian, $Q-P$) and $\hat{v}_I$ for the interference direction (response to a change that carries no driving-relevant information, $N-P$) — unit vectors in representation space, distinct from the behavioural displacements F and I of Section VII, which are the same two constructs read from the output trajectory instead of the activations. A policy reacting along $\hat{v}_F$ is doing what it should; reacting along $\hat{v}_I$ is not: it means the policy moves for a reason that should not move it at all. The two are not merely uncorrelated, they trend opposite one another in aggregate, which is a stronger and more concerning relationship than independence would be.

c) Orthogonality as a trait: We hypothesised that policies whose lighting effect is orthogonal to their pedestrian response would be more stable across domains. The data do not support this. Alignment changes with the driving side for most policies (DiffusionDrive $+0.56 \rightarrow +0.09$, DiffusionDriveV2 $+0.63 \rightarrow +0.16$). Only AutoVLA keeps it ($+0.65 \rightarrow +0.62$). The purer policies shifted more, not less (Table VI, P6). The magnitude of the lighting effect is a trait that a small check-up can measure. Its direction is not, until hundreds of frames are available (Section VI).

IX. Discussion

a) Using the report: Read one exam at a time, the report answers separate questions that a single score merges. Whether a policy is safe because it yields or because it stays short is answered by exposure together with hazard sensitivity: DiffusionDrive keeps its distance by planning short trajectories, which holds only as long as the trajectories stay short, whereas AutoVLA and Alpamayo-1.5 cover a human-like distance without the response to match. Whether a plan that moves is a plan that yields is answered by the decomposition of Section VII: across the 246 NAVSIM scenes and six policies, only 2.4% of cells meet all three conditions, and displacement alone would have ranked the least safe policy first. The exam that no current benchmark reports is scaling: whether the response grows when the hazard does. Five policies fail it, and the one that passes, DiffusionDriveV2, does so at a sensitivity below 0.15.

b) Limitations: The exams are open-loop. The approach sequences are replayed from logs and re-anchored to the logged pose. The night is a synthetic rendering that also halves the number of other pedestrians the detector finds. The counterfactual pairs are constructed by editing, not captured, and so carry noise: 8.5% of removals leave a detectable residue, the inpainting artefact is 5–17% of the effect, and the VLA models can be edited only through their front camera. Alpamayo-1.5 covers 170 of the 236 frames. SimLingo plans 2.5 s; for NAVSIM's 4 s we extrapolate at constant velocity (holding position instead gives EPDMS 0.47, still the lowest). Cross-policy correlations rest on six policies.

c) Future work: The check-up is designed as the first half of a benchmark-to-deployment pairing. The next step is to collect paired data rather than construct it: the same scene with and without a pedestrian, by day and by night. We then plan to validate the diagnoses against the same policies' behaviour on a real vehicle. The sample-size laws above tell us how large that collection must be. A few dozen pairs suffice for planned distance, over-reaction and lighting; hundreds of pairs with a genuine need to react are needed for hazard sensitivity.

X. Conclusion

A benchmark score reports an outcome, not a cause. We read the behaviour that a deployment decision needs from a few hundred logged frames, using counterfactual exams that an independent detector verifies. Three results are worth carrying forward. A near-zero counterfactual response is a reliable certificate that the pedestrian did not enter the decision, while a large response certifies nothing: of the plans that moved, as many moved towards the pedestrian as away from it. What transfers between driving sides is how steadily a policy drives rather than how strongly it reacts, and that exam is the only one—among ours and the standard scores alike—that predicts behaviour on the new side at significance. And each verdict can be priced in advance: some are settled in a dozen frames, others need hundreds, which is what a small check-up must know about itself before it is used to decide anything.

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